A living twin of your theme park.

Connect the park. Understand the ripple effects. Coordinate a better day.

OPERATIONS SIMULATION GUEST EXPERIENCE



ONE PARK · MANY CONNECTED SYSTEMS

Illustrative park concept; no real park data.

A park is a city, a factory and a hospitality business.

A shared capacity model connects guest demand to the assets, people and services that can fulfill it.

Guest experience

Shorter waits. Clearer guidance. Accessible routes. More time enjoying the park.

Operational resilience

Faster incident understanding. Coordinated recovery. Fewer delayed openings.

Asset productivity

Maintenance informed by condition, parts availability and operational impact.

Resource efficiency

Staff, energy, water and food aligned with actual demand.

THE VALUE COMES FROM CONNECTING CONSEQUENCES

Ride closes → Capacity falls

Guests move → Demand shifts

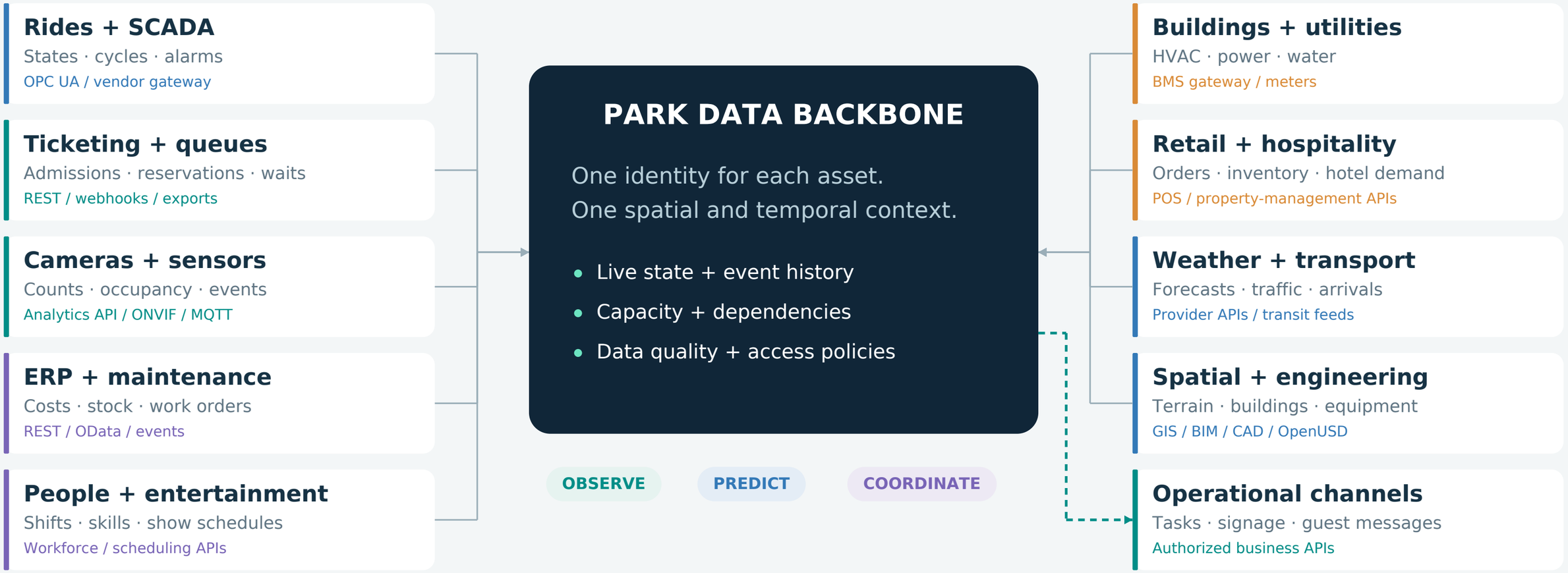
Queues grow → Service pressure

Teams respond → Recovery begins

Success = better guest outcomes + reliable operations + measurable resource savings.

One backbone. Many operational connections.

Adapters normalize identifiers, time and meaning before information reaches the twin.

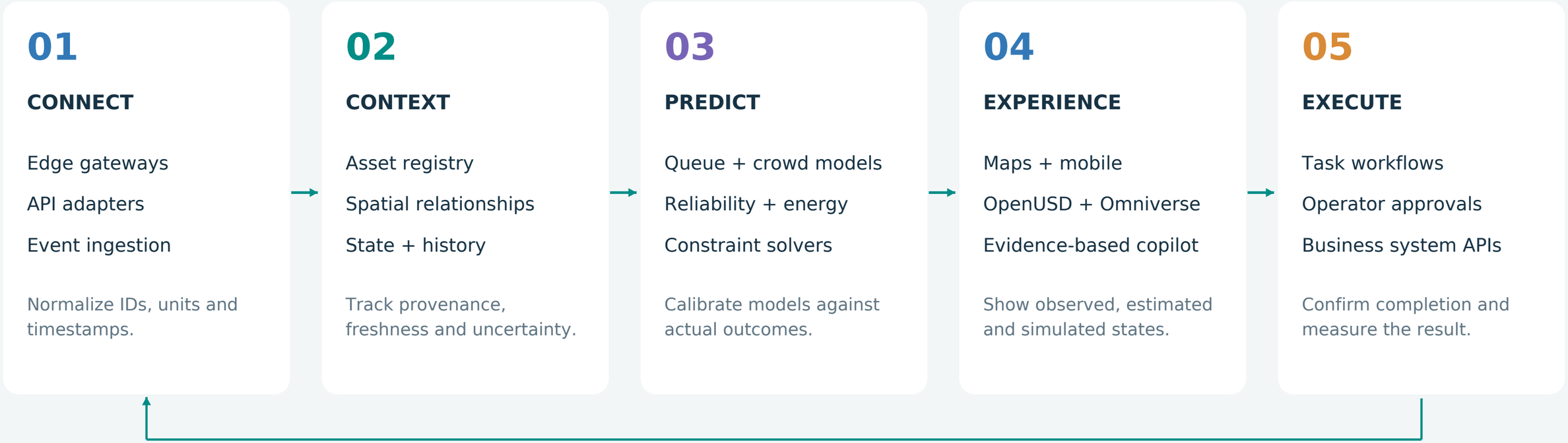


Solid: observations into the twin. Dashed: authorized actions with acknowledgement.

Candidate interfaces; vendor support, access and write permissions require discovery. [4–7]

Separate the data backbone, simulation and experience.

ERP owns business records. Park execution coordinates daily work. The twin connects context across both.



OPERATIONAL DATA

Context API + event history + spatial and telemetry stores. NGSI-LD or a device-twin framework are candidates.

SPATIAL EXPERIENCE

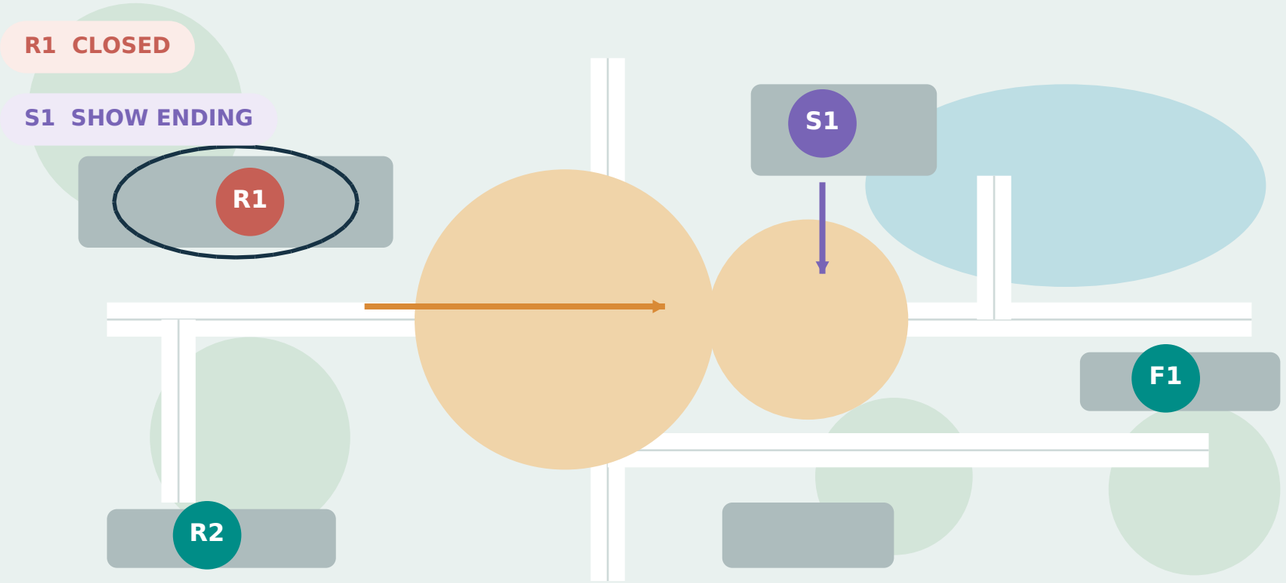
OpenUSD composes the scene. A live adapter binds asset IDs to overlays. Omniverse renders and streams the experience.

Ride safety remains in dedicated local control systems. Proposed architecture. [1–4]

See the park. Follow a dependency. Test a response.

Illustrative control-room concept: the same operational context powers maps, 3D investigation and mobile tasks.

PARK OPERATIONS LIVE REPLAY SCENARIO ILLUSTRATIVE DATA



Ride R1 · Adventure Zone

Observed: unavailable

Source: ride status feed
Last event: 14:10 · illustrative

DEPENDENCIES

Feeder E7 · Queue Q12
Qualified team · Plaza P4

Compare response scenarios

● Observed ● Estimated ● Simulated

3D drill-down: equipment, routes, utilities and sightlines

UI concept only; no live connections. 2D for overview, RTX 3D for spatial investigation. [2–3]

A ride closes just before a nearby show ends.

The twin connects the closure to displaced demand, staffing, guest guidance and recovery.

01 DETECT

Ride status changes.
Check signal freshness.

02 UNDERSTAND

Link queues, show exit, paths and available staff.

03 COMPARE

Simulate demand shifts and feasible responses.

04 COORDINATE

Operator approves tasks and guest messaging.

05 LEARN

Verify execution. Compare forecast and outcome.

THE QUESTION AN OPERATOR CAN ASK

“Which response reduces crowd pressure without creating a new bottleneck?”

Compare practical options

A Reassign qualified staff

B Offer alternate guest routes

C Adjust permitted service capacity

Predictions carry uncertainty.

Scenario timing and asset labels are illustrative. Actions follow existing operating procedures.

Six connected use cases, one reusable foundation.

Prioritize by measurable operational pain, available data and the ability to complete a useful workflow.

01 Guest flow

Balance physical and virtual queues; guide guests using current capacity.

Measure: wait distribution + route success

02 Opening readiness

Combine inspections, staff qualifications, equipment status and access.

Measure: delayed openings + resolution time

03 Maintenance

Link condition, faults, parts and technician availability to park impact.

Measure: downtime + repair duration

04 Energy + water

Coordinate permitted flexible loads; locate leaks and utility dependencies.

Measure: peak demand + use per guest

05 Food + service

Anticipate local demand; dispatch replenishment and cleaning tasks.

Measure: service time + waste + stockouts

06 Design + expansion

Test new attractions, layouts and show schedules before investment.

Measure: capacity gained + bottlenecks

Predictive models need park-specific calibration. Pedestrian simulation reference: [8].

A focused 12-week pilot, with evidence at each gate.

Planning assumption: one themed area, 2–3 attractions, one show venue and access to the core data feeds.

WEEKS 1–2

Discover + baseline

- Map systems and owners
- Confirm API access + latency
- Measure current response

Gate: usable data + agreed KPIs

WEEKS 3–5

Build the shadow

- Create IDs + spatial map
- Connect 4–6 priority feeds
- Show live state + replay

Gate: trusted state + freshness

WEEKS 6–8

Calibrate scenarios

- Model queues + show exits
- Replay historical incidents
- Validate against held-out days

Gate: acceptable forecast error

WEEKS 9–12

Trial + evaluate

- Run in advisory mode
- Trial approved workflows
- Compare outcomes + costs

Gate: evidence to expand or stop

FIRST WEEK

Name the operations sponsor. Select the incident. Inventory the feeds. Agree how success will be measured.

CORE TEAM

Park operations + integration/data engineer + twin/visualization engineer + simulation specialist. Involve IT/OT and ride vendors.

Indicative sequence, not a delivery commitment. Vendor access and data quality may change the schedule.

Fund a measurable operating improvement.

Agree the baseline and acceptance thresholds before building. Expand only when the pilot produces useful evidence.

PILOT SCORECARD

Guest outcome	Wait-time distribution; crowd pressure; accessibility
Operations	Detection-to-action time; task completion; recovery
Model quality	Forecast error; uncertainty calibration; stale data
Adoption	Operator use; accepted recommendations; trust
Economics	Avoided disruption + resource savings − recurring cost

COSTS TO MAKE VISIBLE

- Integration + data cleanup
- Model creation + calibration
- GPU rendering + compute
- Software + operations
- Training + process change

RECOMMENDED NEXT STEP

Authorize discovery for one zone and one incident workflow. Use the evidence to choose the next investment.

No ROI or performance improvement is claimed. Compare matched operating conditions and avoid double-counting benefits.

Reference architecture + a park-specific design proposal.

Based on NVIDIA FOX and supporting platform documentation. This is a concept deck, not a validated park deployment.

1 **NVIDIA Factory Operations (FOX)**

Agent coordination and operational-twin pattern. Article links to an availability notification page.

2 **OpenUSD: Introduction**

Scene composition, references, layers and time-varying properties.

3 **NVIDIA Omniverse: rendering + streaming**

RTX Real-Time 2.0; streaming documentation points new deployments toward NVCF.

4 **FIWARE: Digital Twins**

NGSI-LD context integration; a candidate for the shared park model.

5 **OPC Foundation: OPC UA**

Industrial information exchange and information modeling.

6 **IBM Maximo: Integration framework**

Business objects and REST integration for maintenance workflows.

7 **ONVIF: Profile M**

Interfaces for analytics metadata and events.

8 **AnyLogic: Pedestrian Library**

Queue, density and service-point modeling, including amusement parks.

TO CONFIRM IN DISCOVERY • Vendor access and licensing • Geometry and asset IDs • Data latency and quality
Operating constraints • Model acceptance criteria • Ownership and cost of ongoing maintenance

Sources reviewed 15 October 2026. Reference entries are clickable; full links in accompanying sources file.